

Stochastic Strategic Demand-Capacity Balancing Using Model-Free Reinforcement Learning

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Abstract—Strategic demand-capacity balancing is essential for mitigating inefficiencies in air traffic operations, particularly under dynamic and uncertain weather conditions. This work proposes a stochastic demand-capacity balancing approach based on reinforcement learning, in which an agent allocates ground delays to individual flights using traffic demand and weather forecasts. The method is evaluated using real-world data for São Paulo/Guarulhos International Airport (SBGR), with performance assessed against observed weather conditions. An additional case study illustrates the applicability of the approach to Urban Air Mobility (UAM) operations, showing that the same algorithm can support traffic flow management in both conventional and emerging air traffic environments. The results highlight the potential of reinforcement learning to support adaptive, uncertainty-aware demand-capacity balancing in future Air Traffic Management (ATM) systems.

Index Terms—air traffic management, demand-capacity balancing, reinforcement learning, weather uncertainty, urban air mobility

I. INTRODUCTION

Air Traffic Management (ATM) increasingly relies on data-driven decision-support tools to manage growing demand, capacity constraints, and emerging airspace users. Strategic demand-capacity balancing (DCB) mitigates flight inefficiencies by proactively aligning traffic demand with available capacity, particularly under weather-related operational constraints. The ground-holding problem (GHP), deciding which aircraft to delay on the ground to avoid expensive airborne delays, has motivated decades of optimization research [1]. While strategic DCB allocates ground delays over planning horizons, complementary tactical DCB approaches [2] optimize runway configuration and service rates at operational timescales. However, traditional approaches rely on deterministic forecasts and predefined schemes, limiting adaptability to stochastic demand-capacity variations.

Weather-related delays impose substantial global costs. In the U.S., weather accounts for 74.3% of system-impacting delays exceeding 15 minutes (2017–2023) [3], while European airspace experiences 20–50% weather-attributed Air Traffic Flow Management (ATFM) delays [4]. São Paulo/Guarulhos International Airport (SBGR) in Brazil, processing more than 300,000 annual operations [5], exemplifies major hubs where

weather-induced capacity reductions trigger widespread delays.

Urban Air Mobility (UAM) operations, projected at 55,000 or more daily metropolitan operations in the near term over the U.S. [6], introduce different challenges: short flights, distributed vertiports, tight aircraft turnaround, limited parking, and short-notice service expectations. Strategic planning can significantly reduce UAM conflicts and airborne delays [7], establishing high-density UAM as a critical application domain for adaptive demand-capacity balancing where traditional reactive approaches may become overwhelmed.

This work proposes a model-free reinforcement learning (RL) framework integrating strategic ground-delay allocation with tactical runway configuration management under weather uncertainty. The RL agent learns to allocate delays based on traffic demand and Terminal Aerodrome Forecast (TAF) information, incorporates weather dynamics through stochastic scenario generation, and builds on the tactical framework of Carvalho and Murça [2]. Evaluation on real-world SBGR data (tested against Meteorological Aerodrome Reports) demonstrates effectiveness in conventional operations. The same algorithm is then applied to a UAM case study without modification, illustrating cross-domain applicability. The results highlight reinforcement learning’s potential for adaptive, uncertainty-aware demand-capacity balancing across diverse ATM environments.

II. LITERATURE REVIEW

The ground-holding problem (GHP) has been extensively studied since early work by Richetta and Odoni [1], with Bertsimas et al. [8] providing a foundational integer optimization model for large-scale ATFM covering departure, en route, and arrival phases. However, these deterministic approaches rely on forecasts that may not capture weather uncertainty, operate primarily at strategic levels without tactical runway management integration, and fail to model the non-linear interactions between demand, capacity, and weather characteristic of real operations.

Recognizing these limitations, researchers have developed stochastic formulations. Shone et al. [9] comprehensively review stochastic modeling in ATM, covering queueing theory, robust optimization, and stochastic integer programming. Estes and Ball [10] demonstrated polyhedral properties of stochastic ground-holding models enabling LP relaxations, while Chen

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et al. [11] proposed chance-constrained optimization using Bernstein polynomials for efficient probabilistic constraint solving. A recent bibliometric review by Aditya et al. [12] synthesizes over 160 studies on ATFM optimization (2014–2024), highlighting the critical importance of integrating optimization with real-time data and machine learning for managing dynamic uncertainty.

For UAM, Lee et al. [13] proposed a strategic DCB algorithm for vertiport capacity management and evaluated how scheduling outcomes vary with vertiport capacity, slot-size constraints, operator policies, cruise speed, and departure-arrival time uncertainty.

Adaptive control frameworks have emerged to address weather-induced capacity variations. Jacquillat et al. [14] demonstrated potential benefits of adaptive runway configuration strategies at John F. Kennedy International Airport (JFK) through Markov chain modeling, motivating data-driven and learning-based approaches for real-time uncertainty management. The broader landscape of reinforcement learning in aviation, surveyed by Razzaghi et al. [15], documents RL applications across collision avoidance, air traffic flow management, and aircraft control. While RL shows promise, key challenges remain in validation, certification, and sim-to-real transfer for safety-critical systems. Notably, existing RL approaches for ATFM use multi-agent frameworks, hierarchical planning, and RL-optimization integration, yet none explicitly address integrated strategic-tactical demand-capacity balancing under weather uncertainty with application to both conventional ATM and UAM case studies.

Recent work by Jones et al. [16] and Taylor et al. [17] demonstrated learning-based approaches for recommending traffic management initiatives (TMIs) under forecast uncertainty, with the latter validated over 92 days at Atlanta airport. Most recently, Carvalho and Murça [2] proposed a model-free RL framework for tactical runway capacity management at SBGR. However, existing work primarily focuses on conventional ATM with limited strategic-tactical integration, leaving open the challenge of unified demand-capacity balancing that jointly optimizes strategic ground delays and tactical runway configuration management under weather uncertainty while supporting application across conventional and emerging UAM contexts. This study addresses these gaps by developing an integrated RL framework and applying it to both conventional ATM and a UAM case study without algorithmic modification.

III. METHODOLOGY

This section details the proposed reinforcement learning framework for strategic demand-capacity balancing under weather uncertainty, including problem formulation, capacity modeling for both conventional ATM and UAM, weather uncertainty representation, reward structure, and training protocol.

A. Problem Formulation

Strategic demand-capacity balancing is formulated as a Markov Decision Process (MDP) where the controller operates at 15-minute decision epochs. The formulation is based on the same structure of the runway capacity management problem of Carvalho and Murça [2], allowing the integration of strategic and tactical decisions. At each epoch t , the state comprises:

$$s_t = (RC_{t-1}, a_t, d_t, wc_t, ws_t), \quad (1)$$

where:

- RC_{t-1} = runway configuration active during the previous decision epoch $t - 1$ (A, B, or unavailable),
- a_t = arrival queue length (cumulative unserved arrivals),
- d_t = departure queue length (cumulative unserved departures),
- wc_t = weather condition (VMC, IMC, or LIMC),
- ws_t = wind state (0–3, indicating which configurations are available based on crosswind/tailwind limits).

At each decision epoch, the RL agent selects a strategic action for each arrival flight f , a ground delay $g_f \in \{0, 1, 2, \dots, G_{\max}\}$ representing the number of 15-minute periods to delay it, which reallocates demand across time periods. The value of $G_{\max}$ is limited to one hour ($G_{\max} \leq 4$), and also to the remaining time in the planning horizon, to keep schedule-level demand reallocation operationally moderate. This delay directly shifts the scheduled arrival time from epoch t to epoch $t + g_f$, modifying future queues:

$$a_{t+g_f}^{\text{sched}} \leftarrow a_{t+g_f}^{\text{sched}} + 1, \quad (2)$$

$$a_t^{\text{sched}} \leftarrow a_t^{\text{sched}} - 1, \quad (3)$$

where the reallocation affects subsequent tactical decisions through updated queue states, with a_t^{sched} representing the scheduled arrival demand at epoch t .

For each selected ground delay, the tactical optimization layer then determines the runway configuration from the set of operationally valid options as a function of the wind state:

$$RC_t \in \mathcal{RC}(ws_t), \quad (4)$$

where feasibility depends on crosswind and tailwind limits, considering a maximum of 20 knots for crosswind and 5 knots for tailwind [14].

Given the selected configuration and current weather condition, a tactical optimization layer determines the final service rates for arrivals u_t and departures v_t based on the following heuristic:

$$u_t = \min(u_{\max}[wc_t, RC_t], a_t), \quad (5)$$

$$v_t = \min(v_{\max}[wc_t, RC_t], d_t), \quad (6)$$

where $u_{\max}$ and $v_{\max}$ are the maximum service rates for arrivals and departures under the given weather condition and runway configuration, derived from historical capacity data. This procedure is represented in Figure 1.

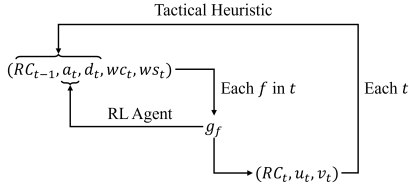


Fig. 1. Interaction between strategic ground delay decisions and tactical runway configuration management.

B. Capacity Modeling

This section details the capacity modeling approaches for both conventional ATM and UAM operations, highlighting the differences in capacity curve structures and the heuristic adaptation for tactical service rate determination.

1) *Airport Capacity Modeling*: Runway configuration capacity envelopes (RCCE) are derived from historical arrival-departure data using quantile regression for conventional traffic [18]. For data points (x_n, y_n) , $n \in \{1, 2, \dots, N\}$ representing arrivals and departures, the linear program minimizes deviations from the capacity frontier:

$$\min \sum_{n=1}^N Z_n, \quad (7)$$

subject to

$$\begin{aligned} Z_n &\geq y_n - (\alpha_k + \beta_k x_n), & k-1 \leq x_n \leq k, \forall n, \\ Z_n &\geq \omega_p (\alpha_k + \beta_k x_n - y_n), & k-1 \leq x_n \leq k, \forall n, \\ \beta_{k+1} &\leq \beta_k, & \forall k \in \{1, \dots, x_{max} - 1\}, \\ \beta_1 &\leq 0, \\ \alpha_k + \beta_k k &= \alpha_{k+1} + \beta_{k+1} k, & \forall k \in \{1, \dots, x_{max} - 1\}. \end{aligned} \quad (8)$$

Here, Z_n represents the deviation of point n from the quantile regression line; α_k and β_k are line parameters for arrival count $x_k = k \in \{1, \dots, x_{max}\}$; and $\omega_p = \frac{100-p}{p}$ is a weighting function for the adopted percentile p . Separate RCCE are constructed for each weather condition: visual meteorological condition (VMC), instrument meteorological condition (IMC), and low instrument meteorological condition (LIMC), by filtering data according to ceiling and visibility thresholds.

The state transition follows:

$$a_{t+1} = a_t + d_t^{\text{sched}} - u_t, \quad (9)$$

$$d_{t+1} = d_t + d_t^{\text{sched}} - v_t, \quad (10)$$

where a_t^{sched} and d_t^{sched} are the scheduled arrivals and departures for epoch t .

2) *Vertiport Capacity Modeling and Heuristic Adaptation*: Unlike conventional ATM with bijective and invertible airport capacity curves, Urban Air Mobility (UAM) operations generate fundamentally different, non-invertible vertiport capacity curves reflecting standardized eVTOL operational procedures [19], [20]. Vertiport capacity is represented as feasible arrival-departure pairs (u, v) indexed by arrival level. For each weather condition wc and runway configuration rc , three arrays define the capacity envelope:

- $\mathbf{u}[wc, rc] = \{0, 1, 2, \dots, u_{\max}[wc, rc]\}$: discrete arrival capacity points,
- $\mathbf{v}_{\min}[wc, rc]$: array where element i gives the minimum departures required if i arrivals are served,
- $\mathbf{v}_{\max}[wc, rc]$: array where element i gives the maximum departures possible if i arrivals are served.

Differently from conventional airport operations, where for each arrival count the minimum number of departures is always zero, vertiport operations require a minimum number of departures to serve a given arrival level due to aircraft availability and limited parking positions. This results in non-invertible capacity curves where the feasible departure range for each arrival level is bounded by both minimum and maximum values, reflecting the operational constraints of eVTOL services.

The capacity envelopes are systematically generated through a parametric constraint-satisfaction approach. The vertiport operates with two distinct operational sides perpendicular to existing runway infrastructure to ensure traffic separation and safety, with each side having different capacity characteristics:

- Side A: Optimized for arrivals (higher maximum arrivals, lower maximum departures),
- Side B: Optimized for departures (lower maximum arrivals, higher maximum departures).

This configuration allows flexible adaptation to demand variations, with weather conditions modulating the capacity of both sides. In the implementation, each weather condition $wc \in \{\text{VMC}, \text{IMC}, \text{LIMC}\}$ and operational side $s \in \{A, B\}$ is represented by integer feasible arrival-departure pairs (u, v) satisfying

$$\begin{aligned} 0 &\leq u \leq U_{\max}[wc, s], \\ 0 &\leq v \leq V_{\max}[wc, s], \\ u + v &\leq C_{\max}[wc, s], \\ u + d_{\min}[wc, s] &\leq v \leq u + d_{\max}[wc, s]. \end{aligned} \quad (11)$$

The parameters are weather-dependent: VMC uses the largest total capacity, while IMC and LIMC progressively reduce $U_{\max}$, $V_{\max}$, and $C_{\max}$. Side A is arrival-favoring and side B is departure-favoring; for example, in VMC the arrival/departure bounds are (9, 7) for side A and (7, 9) for side B. The diagonal constraints encode the need to balance arrivals with departures, representing aircraft availability, parking, and turnaround limitations. The feasible region is obtained by enumerating all integer pairs and retaining those satisfying the constraints. For each feasible arrival count u , the implementation stores the minimum and maximum feasible departure counts, producing arrays $\mathbf{u}[wc, s]$, $\mathbf{v}_{\min}[wc, s]$, and $\mathbf{v}_{\max}[wc, s]$.

The tactical layer first selects the operational side. If the previous side remains feasible under the current wind state, it is retained; otherwise, the available side with the largest maximum arrival capacity is selected. Given the selected side s_t , weather condition wc_t , arrival queue a_t , and departure

queue d_t , service rates are computed by a discrete feasibility search:

$$u_t = \max \{u : u \leq a_t, u \leq \max \mathbf{u}[wc_t, s_t] \wedge \mathbf{v}_{\min}[wc_t, s_t, u] \leq d_t\}, \quad (12)$$

$$v_t = \min(d_t, \mathbf{v}_{\max}[wc_t, s_t, u_t]). \quad (13)$$

Thus, the heuristic searches from the largest candidate arrival count downward until the departure queue is sufficient to meet the required minimum departures. Once u_t is fixed, departures are served up to the corresponding maximum feasible departure count and the available departure queue.

C. Weather Uncertainty Modeling

Weather is represented at 15-minute resolution using three weather conditions: VMC, IMC, and LIMC. Transition probabilities are estimated from historically observed weather conditions from Meteorological Aerodrome Reports (METAR) sequences:

$$P(\text{METAR}_t = j \mid \text{METAR}_{t-1} = i) = \frac{\#\text{transitions } i \rightarrow j}{\#\text{occurrences of } i}. \quad (14)$$

Wind states are derived from wind speed and direction data, translated into crosswind and tailwind components relative to runway orientation. The wind state ws_t is defined as a discrete variable indicating which runway configurations are available based on the following thresholds:

$$ws_t = \begin{cases} 0 & \text{if both available (CW} \leq 20 \text{ and TW} \leq 5) \\ 1 & \text{if only Side A/10R-10L available} \\ 2 & \text{if only Side B/28L-28R available} \\ 3 & \text{if none available} \end{cases}, \quad (15)$$

where CW is crosswind magnitude and TW is tailwind.

Forecast error is captured through confusion matrices estimated by comparing weather forecasts from Terminal Aerodrome Forecasts (TAF) against observed weather (METAR) realizations:

$$P(\text{METAR} = j \mid \text{TAF} = i) = \frac{\#\{\text{TAF} = i, \text{METAR} = j\}}{\#\{\text{TAF} = i\}}. \quad (16)$$

During training, weather and wind states evolve stochastically by sampling from the total likelihood, a product of transition and confusion probabilities, capturing both temporal weather dynamics and forecast uncertainty:

$$wc_t \sim P(wc_t \mid wc_{t-1}, \widehat{wc}_t) \propto P(wc_t \mid wc_{t-1}) \cdot P(wc_t \mid \widehat{wc}_t), \quad (17)$$

$$ws_t \sim P(ws_t \mid ws_{t-1}, \widehat{ws}_t) \propto P(ws_t \mid ws_{t-1}) \cdot P(ws_t \mid \widehat{ws}_t), \quad (18)$$

where $P(wc_t \mid wc_{t-1})$ and $P(ws_t \mid ws_{t-1})$ are transition probabilities estimated from historical weather observations (METAR), and $P(wc_t \mid \widehat{wc}_t)$ and $P(ws_t \mid \widehat{ws}_t)$ are confusion probabilities capturing forecast (TAF) prediction errors [2]. This stochastic scenario generation allows the RL agent to

learn policies that are robust to weather uncertainty, as it encounters a wide range of plausible weather evolutions during training.

D. Reward Function

The cumulative cost minimized across an episode decomposes into tactical and strategic components that are dynamically evaluated as flights are processed:

$$\mathcal{J} = \mathcal{J}^{\text{tac}} + \mathcal{J}^{\text{strat}}, \quad (19)$$

The tactical cost component addresses runway capacity management at each time step t , evaluated for all remaining flights given the current queue state by the following composition between queue penalties and configuration-switching costs:

$$\mathcal{J}^{\text{tac}} = \mathcal{J}^{\text{queue}} + \mathcal{J}^{\text{switch}}, \quad (20)$$

which can be expressed as:

$$\mathcal{J}^{\text{tac}} = \sum_{t=0}^T [\alpha(a_t - u_t)^2 + (d_t - v_t)^2 + \delta \mathbb{I}(RC_t \neq RC_{t-1}) \cdot (\alpha a_t^2 + d_t^2)], \quad (21)$$

where α is the relative cost of airborne delays compared to ground delays, δ is the relative cost of switching runway configurations, and a_t and d_t represent the queue lengths at the beginning of time step t given current demand and runway management decisions.

When a strategic ground delay g_f is applied to flight f at time t , the flight's demand is reallocated from epoch t to epoch $t + g_f$. This reallocation modifies all future queue states, changing the tactical cost:

$$\Delta \mathcal{J}^{\text{tac}} = \mathcal{J}_{\text{delayed}}^{\text{tac}} - \mathcal{J}_{\text{original}}^{\text{tac}}, \quad (22)$$

capturing the impact of demand redistribution on runway configuration decisions and queue penalties over the remaining episode.

The strategic cost component reflects the direct penalty for ground delay decisions on individual flights:

$$\mathcal{J}^{\text{strat}} = \eta \sum_{f=1}^F g_f, \quad (23)$$

where:

- F is the total number of flights in the episode,
- g_f is the ground delay (in 15-minute periods) assigned to flight f ,
- $\eta > 0$ is the weight for strategic delay costs (adopted $\eta = 0.1$).

The total reward at each step combines the change in tactical cost due to flight reallocation with the immediate strategic penalty:

$$R_t = - \left(\Delta \mathcal{J}^{\text{tac}}(t) + \eta \sum_{f=1}^{F_t} g_f \right), \quad (24)$$

where F_t is the total number of flights for which ground delay decisions are made at time step t . The reward structure

incentivizes the agent to learn strategic delay allocations that minimize overall costs by considering both the immediate penalty of ground delays and their downstream impact on tactical runway management and queue conditions.

This design ensures the agent learns to balance strategic ground delays against their tactical consequences, as demand redistribution may improve queue conditions at certain future time steps while deteriorating them at others. The agent’s objective is to learn a policy $\pi_\theta(a | s)$ that maximizes the expected cumulative reward over episodes, effectively learning to allocate ground delays in a way that optimizes both strategic and tactical outcomes under weather uncertainty.

E. Training Protocol

The policy $\pi_\theta(a | s)$ is learned using Maskable Proximal Policy Optimization (PPO) [21] from Stable Baselines 3 [22]. The environment was implemented as a Gymnasium environment [23]. At each step, the agent receives a binary mask $m_t \in \{0, 1\}^{|A|}$ indicating valid actions:

$$m_t[a] = \begin{cases} 1 & \text{if } a \in \{0, 1, \dots, \min(G_{\max}, T - t)\} \\ 0 & \text{otherwise} \end{cases}, \quad (25)$$

where $T = 96$ is the number of 15-minute periods in the daily planning horizon.

The masked policy becomes:

$$\tilde{\pi}_\theta(a | s) = \frac{\pi_\theta(a | s) \cdot m_t[a]}{\sum_{a'} \pi_\theta(a' | s) \cdot m_t[a']}, \quad (26)$$

ensuring the agent never selects infeasible ground-delay actions that would move a flight beyond the remaining planning horizon. Runway configuration is not selected by the RL action; it is determined afterward by the tactical layer from the operationally valid options. This significantly improves sample efficiency and accelerates convergence.

Training is conducted over 300,000 episodes for UAM and 500,000 for conventional ATM for each of the 30 randomly selected days, using TAF-based stochastic scenarios for learning, direct TAF forecasts for schedule generation, and METAR observations for testing. Maskable PPO uses stable default values from prior PPO implementations and preliminary convergence checks: learning rate 3×10^{-4} , batch size 64, entropy coefficient 0.01, and clipping ratio 0.2. These values were not exhaustively tuned; sensitivity to RL hyperparameters remains a limitation.

Evaluation compares the learned policy with a no-rescheduling baseline, i.e., the same tactical heuristic and capacity model with $g_f = 0$ for all flights. The simulator is anchored in historical SBGR demand, weather observations, TAF forecast errors, and RCCE capacity envelopes, but it is still an operational abstraction rather than a certified airport digital twin. Delay costs are calibrated from EUROCONTROL data [24]: approximately €45 per delayed departure minute and €90 per airborne arrival minute. In evaluation, the strategic rescheduling cost is omitted so that the reported metric isolates tactical congestion reduction from demand reallocation. The training and testing procedure is illustrated in Figure 2.

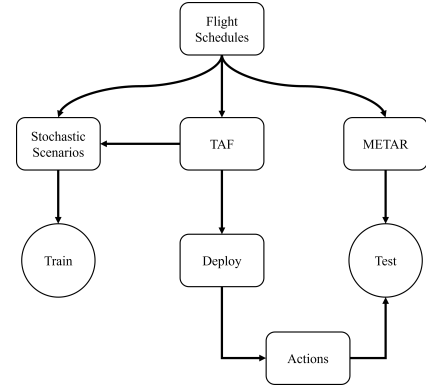


Fig. 2. Training and evaluation protocol for the strategic RL-based demand-capacity balancing framework.

IV. CASE STUDIES

This section describes the datasets used for training and testing the proposed RL framework, including real-world data from SBGR and the synthetic data generation process for UAM scenarios.

A. Real-World Data: SBGR

The primary case study uses data from Guarulhos International Airport (SBGR), a major South American hub that experiences significant congestion and operates under two main runway configurations for arrivals/departures: 10R/10L and 28L/28R. The dataset comprises:

- **Observed Weather Data:** 15-minute meteorological observations (adapted from hourly records) spanning approximately 24 months (2022–2023) from Meteorological Aerodrome Reports (METAR), including wind direction, wind speed, visibility, ceiling, and weather condition indicators (IMC/LIMC flags), obtained from the Meteorological Network database (REDEMET) [25].
- **Weather Forecast Data:** Forecasts of the same information as METAR data at 15-minute resolution (adapted from hourly records) from Terminal Aerodrome Forecasts (TAF), providing forecast information for training the RL agent using the stochastic process, obtained from REDEMET [25].
- **Flight Data:** Scheduled and observed arrival and departure times with 40 NM arrival estimates for all flights at SBGR. Observed flight times are obtained from ANAC’s Active Regular Flight (VRA) database [26] and FlightRadar24’s flight tracks [27], while scheduled flight data is obtained from ANAC’s Scheduled Information System (SIROS) [28]. Observed flight times are used to correct scheduled times (removing buffers), which are used to train and test the RL agent on realistic demand patterns.

From weather observations and predictions, transition and error probabilities are derived. Weather observations (METAR) are used for testing, while forecast data (TAF) is used for training. The dataset is preprocessed to extract relevant features and construct the state space for the RL environment.

Conventional Air Traffic Management (ATM) operations at SBGR provide a baseline for evaluating the proposed strategic

demand-capacity balancing framework. The dataset comprises 30 randomly selected days with diverse weather and demand conditions. Figure 3 illustrates the runway configuration capacity envelopes (RCCE) for SBGR under different weather conditions, derived from historical data using quantile regression [18]. The RCCE reveal significant capacity reductions under adverse weather conditions, with the most pronounced impact observed under low instrument meteorological conditions (LIMC).

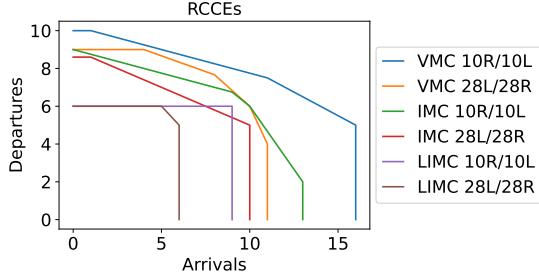


Fig. 3. Runway configuration capacity envelopes (RCCE) for SBGR under different weather conditions [2].

B. Synthetic UAM Data Generation

For the Urban Air Mobility (UAM) case study, a synthetic demand generation pipeline based on actual helicopter traffic distributions obtained from the Air Traffic Movement Information Database (BIMTRA) [29] produces traffic patterns while preserving real weather observations:

- 1) Extract daily and 15-minute slot demand distributions from helicopter traffic data (BIMTRA).
- 2) Scale arrivals and departures independently to create demand levels for the 30 test days. Both scale factors are set to 1.0, preserving the BIMTRA daily demand level while resampling arrival and departure schedules separately.
- 3) Generate individual flight records with realistic flight-duration variability obtained from BlueSky simulations [30] of eVTOL operations destined for SBGR following the current Helicopter Special Routes (REH) and possible vertiport locations [31]. The eVTOL flight profile and technical parameters are based on the study of Sarkar et al. [32], with maximum cruise speed of 96 kt and cruise altitude of 1500 ft AGL with intermediate steps at 300 ft AGL and 48 kt.
- 4) Pair synthetic demand with METAR and TAF data from SBGR to create realistic weather scenarios for training and testing the RL agent in UAM contexts.

This baseline intentionally does not pre-optimize schedules against vertiport capacity; capacity feasibility is evaluated by the tactical layer. The UAM case study is designed as a demanding high-density scenario, consistent with the expectation that mature UAM may exceed current helicopter traffic levels and that traffic-management tools are most valuable under stressed demand. The assumptions are that helicopter temporal patterns provide a first-order proxy for early UAM demand,

SBGR METAR/TAF data represent the local weather affecting the vertiport area, and low-altitude route-level weather is unavailable. Consequently, the case study is not a forecast of mature air-taxi demand, and relocation flights, on-demand request generation without scheduled flight plans, and operator-specific fleet balancing are outside the current scope; on-demand flights would require a tactical decision model.

UAM operations at SBGR represent an emerging operational domain with different capacity characteristics compared to conventional ATM. UAM services employ eVTOL aircraft with shorter flight times, shorter planning horizons, short-notice service expectations, and non-invertible capacity curves reflecting aircraft availability, parking capacity, and turnaround time requirements [19], [20]. The synthetic UAM demand generation pipeline (Section IV-B) combines helicopter traffic distributions from BIMTRA with SBGR weather observations, grounding the scenario in operational plausibility while enabling controlled demand scaling.

Figures 4–7 establish the operational context for UAM at SBGR. Three candidate vertiport locations are identified based on heliport distributions and income patterns [31]. The eVTOL flight profiles conform to realistic operational procedures derived from BlueSky simulations [30], including climb-to-300-ft and cruise-to-1500-ft stages [32]. The hourly demand distribution, extracted from existing helicopter operations, provides a realistic baseline for synthetic flight generation while exposing temporal demand patterns.

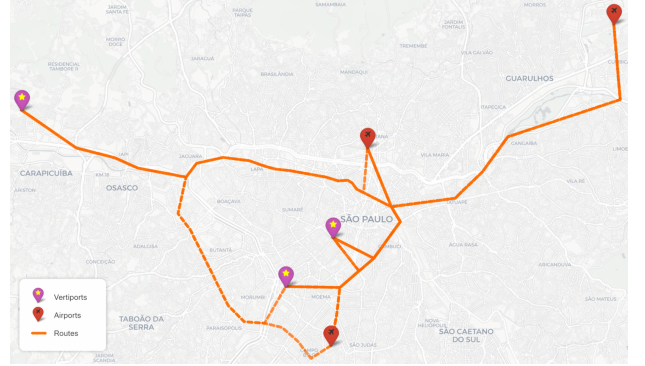


Fig. 4. Three vertiport candidates obtained from the heliport distribution among the São Paulo Metropolitan Region (RMSP), the airports SBGR, SBSP, and SBMT, and the UAM optimal routes connecting them [31].

Figure 8 presents the UAM capacity curves for SBGR under VMC, IMC, and LIMC conditions [19], [20]. Unlike the linear/invertible capacity curves of conventional ATM (Figure 3), UAM capacity envelopes exhibit non-invertible characteristics: for each arrival count, the departure feasible region is bounded by both minimum (reflecting aircraft availability constraints) and maximum (reflecting parking/handling capacity) departure rates. This structural difference fundamentally alters the interplay between strategic and tactical decisions, as reallocation of arrivals may require corresponding adjustments in departures to maintain operational feasibility.



Fig. 5. Example eVTOL flight profile in BlueSky simulator [30] for a route between SBSP and SBGR, showing the flight path and intermediate waypoints of the Helicopter Special Routes (REH).

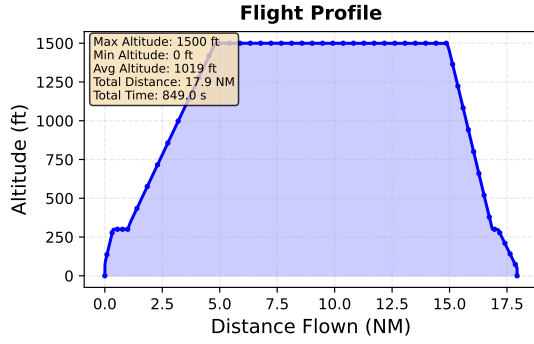


Fig. 6. Altitude (ft) as a function of distance flown (NM) for an eVTOL flight between SBSP and SBGR in the BlueSky simulator [30].

V. RESULTS AND DISCUSSION

This section presents the results of the RL-based strategic demand-capacity balancing policy compared to the baseline tactical policy across the 30 test days at SBGR. The analysis focuses on tactical cost performance, distribution of outcomes, and the relationship between baseline costs and percentage improvements achieved by the learned policy.

A. Case 1: SBGR Conventional Operations

Table I summarizes tactical cost performance across 30 test days for the conventional ATM traffic at SBGR. The RL-based strategic demand-capacity balancing (SDCB) policy achieves a mean tactical cost of €640.4 compared to €738.2 for the baseline, a 13.3% aggregate mean-cost reduction, indicating

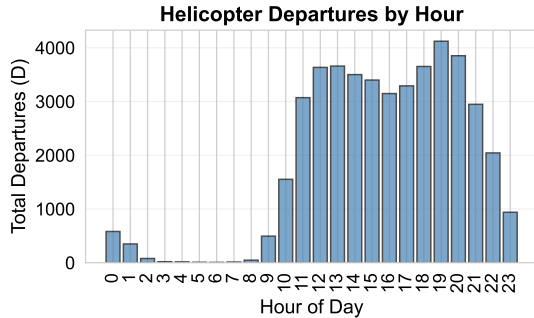


Fig. 7. Hourly demand distribution for helicopter operations at SBGR obtained from the Air Traffic Movement Information Database (BIMTRA) [29].

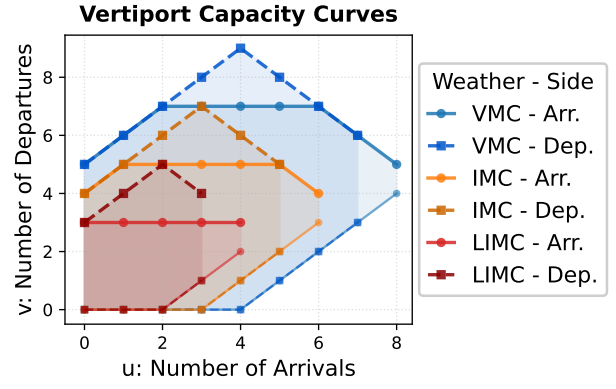


Fig. 8. Capacity curves for SBGR under different weather conditions for UAM operations.

that improvements concentrate in specific congested scenarios. The interquartile range (IQR) for SDCB is €847.1 compared to €1060.9 for baseline, demonstrating lower variability in outcomes.

TABLE I
TACTICAL COST STATISTICS (IN EUR) FOR SDCB AND BASELINE POLICIES ACROSS 30 TEST DAYS AT SBGR.

Metric	SDCB	Baseline
Min	0.0	0.0
Max	2299.5	2529.0
Mean	640.4	738.2
Median	398.3	398.3
Std Dev	692.0	773.7
25th Percentile	94.5	155.3
75th Percentile	941.6	1216.1
IQR	847.1	1060.9

The violin plot in Figure 9 reveals the distribution of outcomes across the 30 test days. The learned policy achieves a 13.3% aggregate mean-cost reduction. The SDCB policy exhibits a narrower distribution concentrated in lower-cost regions, while the baseline shows a broader, higher-cost distribution with longer tails indicating occasional severe congestion scenarios. The Wilcoxon signed-rank test confirms statistical significance ($p\text{-value} = 0.002183$), rejecting the null hypothesis that SDCB and baseline costs are identically distributed. This non-parametric test is appropriate given the non-normal cost distributions.

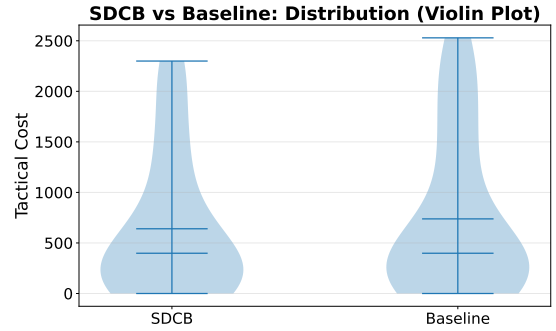


Fig. 9. Performance of the RL-based demand-capacity balancing policy at SBGR across 30 test days.

Figure 10 presents cost reductions sorted by magnitude,

revealing the heterogeneity of algorithmic performance across test days. The results demonstrate consistent improvements on 12 days (40%) with cost reductions exceeding zero, while 18 days (60%) show zero or negligible changes where baseline costs are already minimal. The maximum improvement of 77.2% occurs on high-congestion days where strategic ground delays effectively redistribute demand to less congested periods. Additionally, no day exhibits performance degradation (negative improvement), indicating the algorithm learns robust policies that do not worsen outcomes on any scenario.

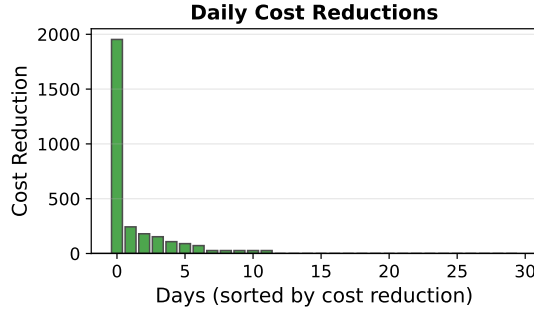


Fig. 10. Cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for the 30 test days sorted by cost reduction.

Figure 11 relates each day's baseline cost to the percentage reduction achieved by SDCB. Figure 12 ranks the same daily percentage reductions from largest to smallest. Bars with nonzero magnitude indicate days improved by SDCB, while near-zero bars indicate days where the baseline was already lightly congested. The top 12 days (40%) range from modest gains ($\approx 5\%$) to 77.2%, showing that SDCB is most valuable when strategic reallocation can move demand away from congested periods.

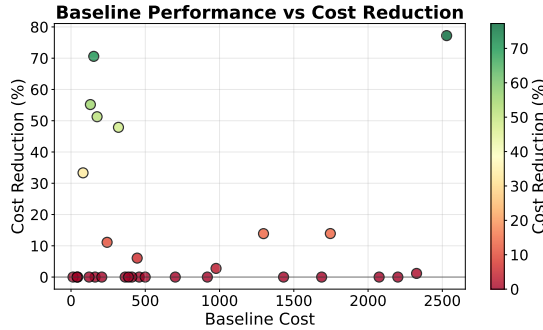


Fig. 11. Percentage cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for each of the 30 test days.

Table II documents the distribution of strategic ground delay decisions across all test flights at SBGR. The RL agent demonstrates conservative delay allocation: 98.9% of flights receive no delay, while only 1.1% are delayed by 1, 2, or 4 time-slot increments (15–60 minutes). This conservative strategy protects airlines from the operational costs of rescheduling flights, concentrating delays on situations where they can have the most significant impact.

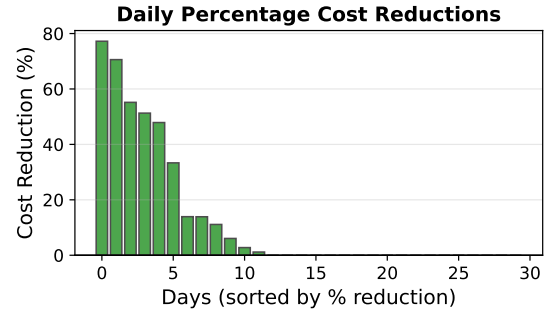


Fig. 12. Percentage cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for the 30 test days sorted by percentage reduction.

TABLE II
DISTRIBUTION OF STRATEGIC GROUND DELAY DECISIONS APPLIED ACROSS ALL TEST FLIGHTS AT SBGR. ONLY 1.1% OF FLIGHTS ARE DELAYED, MOSTLY (1.1%) RECEIVING 1 OR 2 TIME-SLOT DELAYS (15 OR 30 MINUTES).

Time-Slot Delay	Count	Percentage
0	10033	98.9%
1	68	0.7%
2	39	0.4%
3	0	0.0%
4	4	0.0%

B. Case 2: SBGR Urban Air Mobility Operations

Table III reveals improved tactical cost performance for UAM. The SDCB policy achieves a mean tactical cost of €213.2 versus €603.8 for baseline, a 64.7% aggregate mean-cost reduction; median costs fall from €227.3 to €74.3 (67.4%), and the maximum cost from €8730.0 to €1440.0 (83.5%). This aggregate measure differs from the 43.9% mean daily percentage reduction because daily baseline congestion is heterogeneous. These results suggest that the approach can support UAM flight-planning scenarios under the case-study assumptions. For comparability, UAM uses the same representative delay-cost values as conventional ATM, not actual UAM operator costs.

TABLE III
UAM TACTICAL COST STATISTICS (IN EUR) ACROSS 30 TEST DAYS, USING REPRESENTATIVE CONVENTIONAL ATM DELAY COSTS FOR COMPARABILITY.

Metric	SDCB	Baseline
Min	0.0	0.0
Max	1440.0	8730.0
Mean	213.2	603.8
Median	74.3	227.3
Std Dev	319.3	1539.0
25th Percentile	0.0	21.4
75th Percentile	309.4	642.4
IQR	309.4	621.0

The UAM performance distribution (Figure 13) demonstrates superior and more consistent improvements compared to conventional ATM. The learned policy achieves a mean daily percentage cost reduction of 43.9%. The SDCB violin exhibits a tight clustering at low costs with minimal variability, while the baseline shows a broad, high-variance distribution with substantial outliers representing severe congestion episodes. The Wilcoxon signed-rank test (p -value = 0.000281)

confirms highly significant differences between distributions, with substantially stronger evidence than conventional ATM ($p = 0.002183$).

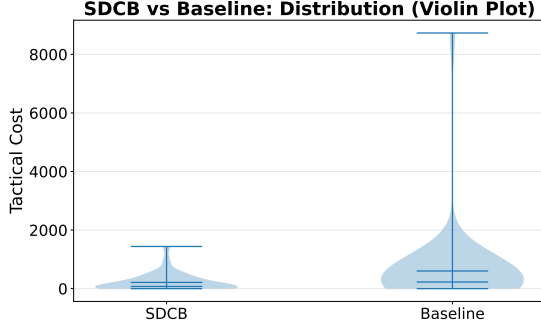


Fig. 13. Performance of the RL-based demand-capacity balancing policy at SBGR across 30 test days.

The UAM reduction plots have the same interpretation: Fig. 14 sorts absolute daily cost reductions, Fig. 15 relates baseline cost to percentage reduction, and Fig. 16 ranks percentage reductions. UAM improves on 23 of 30 days (76.7%), compared to 12 of 30 (40%) for conventional ATM, with several near-100% reductions where strategic delays remove tactical congestion. This occurs partly because conventional flight schedules are already operationally planned, whereas the synthetic UAM baseline is not pre-optimized against vertiport capacity.

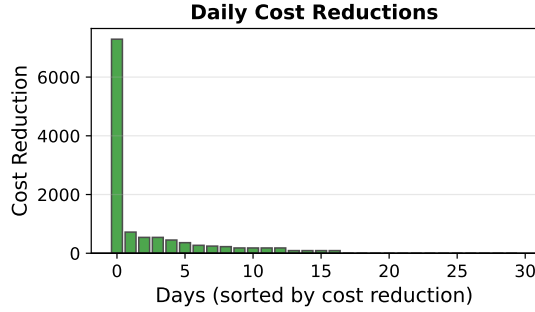


Fig. 14. UAM cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for each of the 30 test days.

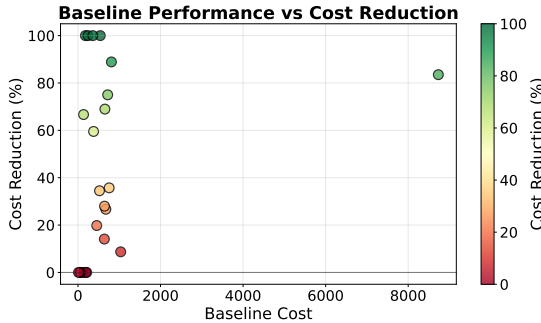


Fig. 15. UAM percentage cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for each of the 30 test days.

Table IV documents UAM ground delay decisions. The policy delays 4.4% of flights, mostly by 1–2 time slots; this

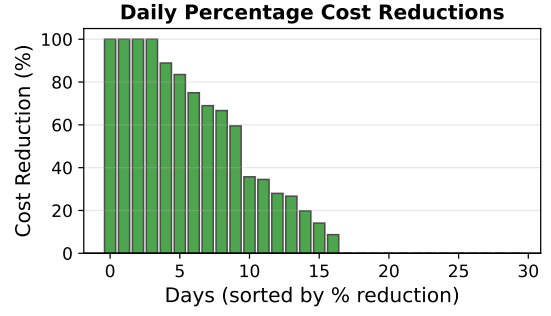


Fig. 16. UAM percentage cost reduction achieved by the RL-based demand-capacity balancing policy compared to the baseline for the 30 test days sorted by percentage reduction.

higher intervention rate, combined with larger cost reductions, suggests strategic delays are particularly cost-effective for UAM operations.

TABLE IV
DISTRIBUTION OF STRATEGIC GROUND DELAY DECISIONS FOR UAM.

Time-Slot Delay	Count	Percentage
0	2731	95.6%
1	62	2.2%
2	50	1.8%
3	5	0.2%
4	9	0.3%

A key result is that the same algorithmic architecture can be applied to both conventional ATM and the UAM case study without modification. The RL framework automatically specializes through reward signals and learned state-action mappings, discovering that conventional ATM benefits from minimal delays (98.9% no delay) while UAM benefits from moderate strategic intervention (4.4% delayed). This algorithmic robustness suggests the framework can transfer across operational regimes and supports future extension to heterogeneous mixed-fleet scenarios or multi-airport networks.

Several factors contribute to the success of reinforcement learning for stochastic demand-capacity balancing. Unlike rule-based heuristics or traditional optimization methods, the RL agent learns state-dependent strategies that adapt to complex, non-linear interactions between demand, capacity, and weather. Restricting the action space to feasible ground-delay actions via binary masks significantly accelerates learning and eliminates infeasible exploration. By integrating tactical optimization into the reward signal, the strategic policy naturally internalizes the future impact of current decisions. Finally, the same architecture is applied to conventional ATM and the UAM case study without algorithmic modification, suggesting transferability to diverse airport and traffic types.

VI. CONCLUSION

This work contributes a model-free reinforcement learning framework for strategic demand-capacity balancing that handles weather uncertainty and applies to both conventional ATM and UAM operational environments. The stochastic formulation uses probabilistic weather representations and empirically grounded demand scenarios, while action masking encodes

valid ground-delay actions, accelerates learning, and eliminates infeasible actions. Strategic and tactical optimization are integrated by coupling ground delay allocation with configuration and service-rate decisions. Evaluation with 24 months of SBGR METAR/TAF data shows a 13.3% aggregate mean-cost reduction for conventional traffic, with reductions in 12 of 30 days and no performance degradation. The same algorithm applied to UAM traffic management achieves a 43.9% mean daily percentage cost reduction, with improvements in 23 of 30 days.

Current limitations motivate further research. The study is limited to SBGR, so network effects, airline recovery, passenger-connection impacts, multi-airport coordination, and the adaptation of wind-state definitions to other airports remain unexplored. The weather state uses VMC/IMC/LIMC categories and airport METAR/TAF data; route-level low-altitude weather and finer meteorological variables could improve realism. Finally, the UAM case study uses helicopter-based distributions without relocation flights or mature air-taxi demand calibration, so real high-density UAM patterns would require recalibration.

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